

Multiple lines of evidence reveal rapid, seasonal watershed responses to enhanced weathering

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Enhanced weathering accelerates carbon dioxide removal from the atmosphere through the rapid dissolution of land amendments of crushed, cation-rich rocks. The export of enhanced weathering dissolution products to drainage networks has important implications for carbon dioxide removal durability and measurement yet has received little attention. Here we observe rapid and pronounced streamwater chemistry responses within weeks of a crushed basalt application to a temperate headwater catchment: alkalinity exceeded counterfactual estimates by as much as 557 $\mu\text{eq l}^{-1}$, accompanied by changes in silica, lithium and major-ion chemistry consistent with basalt weathering. Difference-in-differences analysis identified repeated treatment effects with enhanced weathering accounting for 7–17% of streamwater alkalinity in successive summers and autumns, but less in winter and spring. This seasonal response reflects a combination of temperature controls on basalt dissolution kinetics and hydrologic controls on weathering-product export. Over 2 years, 9.5–11% of theoretical carbon dioxide removal potential was exported as alkalinity (34.7 t $\text{CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$).

Anthropogenic CO_2 emissions are the primary driver of climate change and ocean acidification, affecting both terrestrial and marine climates and ecosystems^{1,2}. To achieve the Paris Agreement goal of limiting global warming to less than 2 °C, large-scale carbon dioxide removal (CDR) will be necessary³. Terrestrial enhanced weathering (EW) is a CDR approach that involves spreading powdered silicate rocks (for example, basalt) over agricultural and other lands to accelerate chemical weathering reactions that convert atmospheric CO_2 to alkalinity (primarily HCO_3^-) and release base cations^{4,5}. These weathering products may be retained within the watershed or exported with soil waters and

groundwaters to streams and rivers that drain into the ocean, where storage lifetimes may exceed tens of millennia^{6,7}. Recent theoretical assessments indicate that EW implemented on global croplands could sequester 0.5 to 2 Gt $\text{CO}_2 \text{ yr}^{-1}$ at costs comparable to other CDR strategies, which become more competitive when agriculture co-benefits from improved soil health are considered^{8–10}. Despite the growing interest in EW, uncertainty in its practical potential remains and is underpinned by a scarcity of both observations and demonstrations of approaches suitable for quantifying applied-silicate weathering at the field scale.

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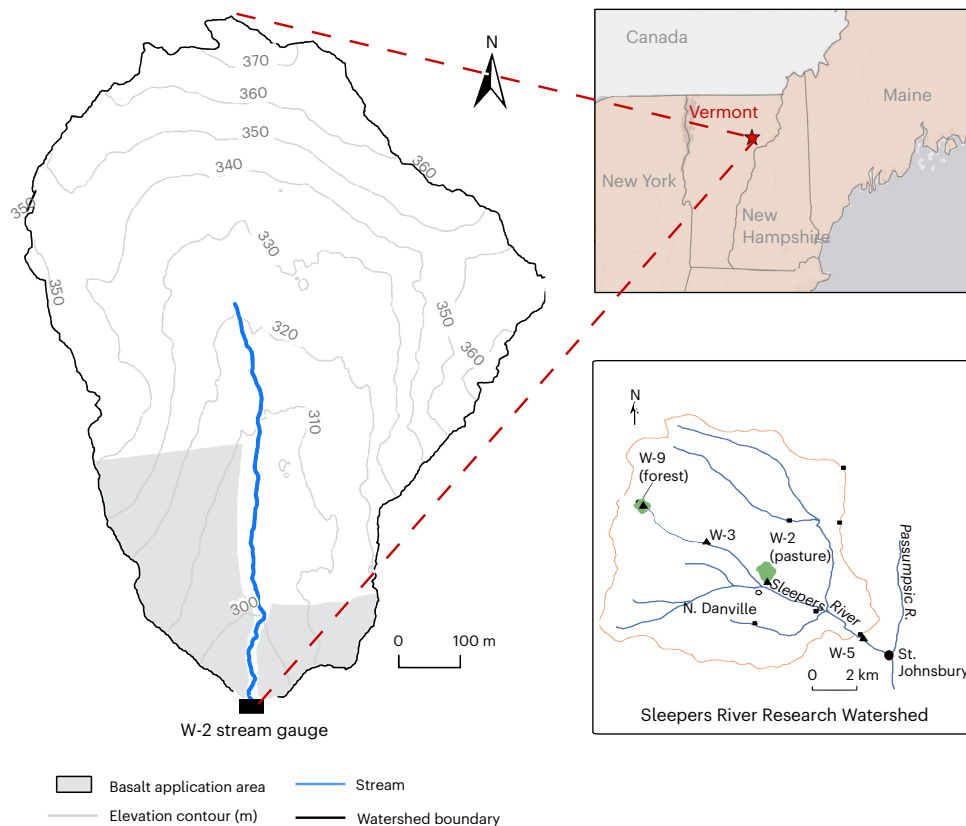


Fig. 1 | Map of the W-2 watershed and basalt application area within the Sleepers River Research Watershed in Vermont, USA. The grey area shows the basalt application area. The lower inset shows the position of W-2 relative to the W-9 reference watershed in the Sleepers River Research Watershed (adapted with permission from ref. 23, Wiley).

Most EW estimates rely on silicate-feedstock dissolution rates determined in laboratories or in near-surface soils from plot-scale or farm-field experiments^{11–13}, thereby neglecting hydrologic and geochemical processes beneath shallow soil horizons that may modify CDR as EW products are transported to surface waters. For example, ion exchange may temporarily retain base cations derived from feedstock dissolution and release protons that neutralize bicarbonate, thus delaying alkalinity generation and its export to surface water until those cations are later desorbed¹⁴. Secondary-clay formation and carbonate precipitation in groundwaters beneath application areas may sequester feedstock weathering products, effectively reducing net CDR^{7,15–17}. Moreover, the routing of rainfall along watershed flow paths will introduce time-varying hydrologic lags between feedstock dissolution in near-surface soils and the expression of alkalinity in streamwater, potentially affecting downstream CDR quantification and underscoring the need for watershed-scale monitoring that captures these transport dynamics.

The streamwater chemical response to EW is likely to be dynamic, shaped by factors that vary at seasonal and sub-seasonal timescales. Seasonal changes in temperature within EW watersheds can affect silicate weathering through abiotic and biotic mechanisms^{18–20}. Weathering rates increase with temperature, a process often modelled following Arrhenius kinetics. Temperature-induced increases in soil $p\text{CO}_2$ (via soil respiration) and organic acids (via root and microbial exudation) will further accelerate dissolution^{20,21}. Beyond temperature, variations in precipitation will modulate watershed wetness and flow-path connectivity that, in turn, govern catchment-water residence times and hence the extent of water-feedstock interactions²². An understanding of how temperature–precipitation interactions influence temporal variation in EW contributions to streamwater alkalinity is essential to informing EW monitoring schemes; however, the ways these seasonal processes regulate alkalinity generation and export at the watershed scale are largely unknown.

Here we present multiple lines of evidence demonstrating rapid, seasonal watershed responses to EW, using high-resolution streamwater chemistry data from a 59-ha hay- and pasture-dominated headwater catchment (W-2) within the Sleepers River Research Watershed^{23–25}, Vermont, USA (Fig. 1). Following a 1-year baseline monitoring period, we applied basalt powder (hereafter ‘basalt’) in June 2023 over 8.9 ha of the southern (downstream) portion of the catchment (15% of the total catchment area) at a rate of 20 t ha^{-1} . We isolated the basalt-weathering signal from background water chemistry by analysing streamwater concentration–discharge ($C-q$) relations before and after basalt application and by comparing streamwater observations to a no-basalt counterfactual estimated from weighted regressions on time, discharge and season (WRTDS)^{26,27}. We then used a generalized additive model coupled with a difference-in-differences (GAM-DiD) approach²⁸ that leveraged observations from a nearby reference watershed (W-9) to quantify seasonal dynamics in the streamwater export of basalt-derived alkalinity, base cations and associated CDR rates. In addition, we applied an endmember mixing analysis to evaluate the effects of basalt treatment on seasonal shifts in weathering-product concentrations within quick-flow and slow-flow pathways that transmit solutes to the stream. This watershed-scale EW study describes how streamwater chemistry responds to the deployment of silicate feedstock and introduces a transferable framework for evaluating the seasonal dynamics of EW-attributable CO_2 removal.

Results and discussion

Rapid and pronounced streamwater chemistry responses to watershed-scale EW

Streamwater alkalinity and base cation concentrations in the W-2 stream responded within 1 month of the basalt application and remained elevated relative to counterfactual WRTDS estimates from

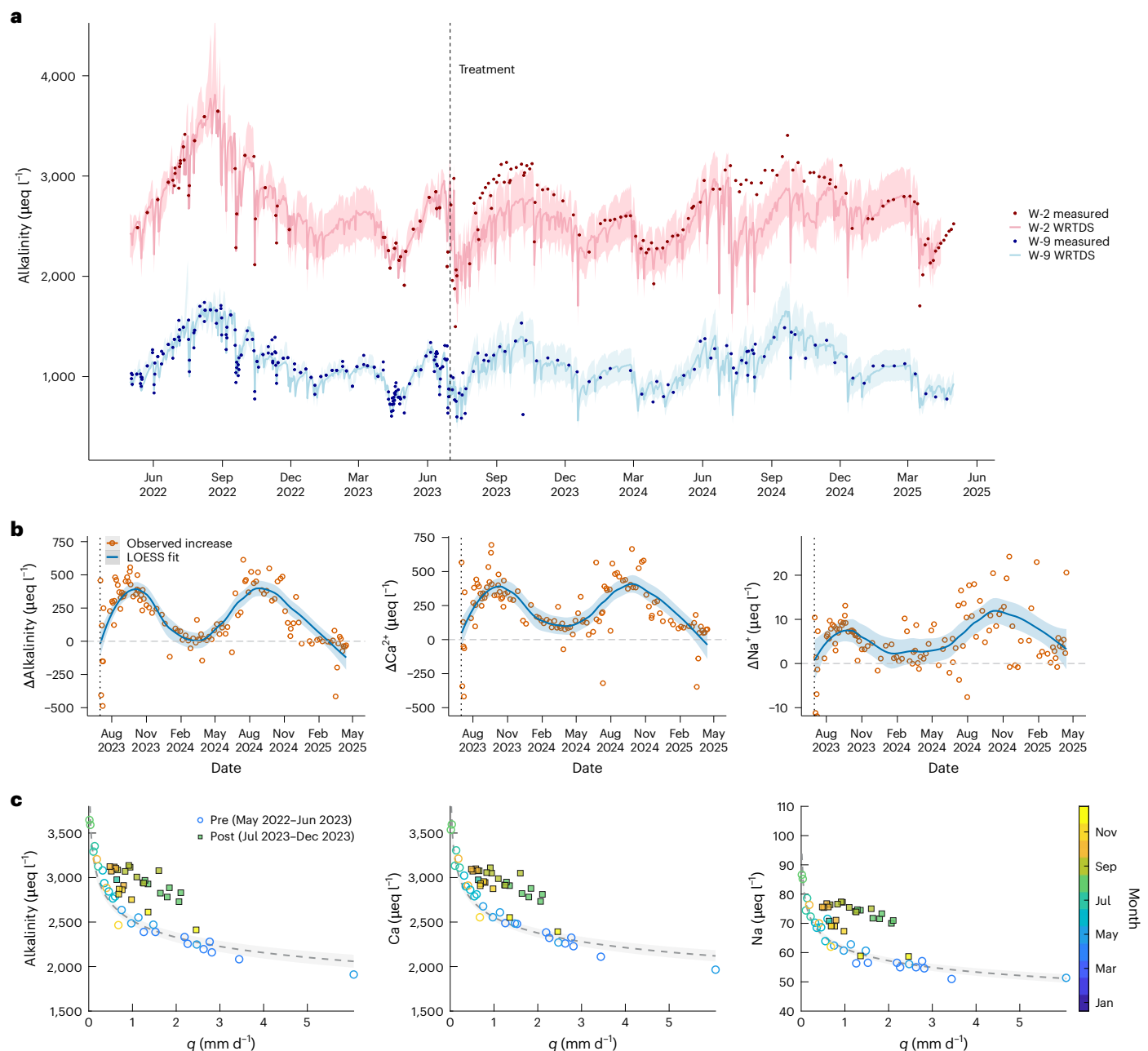


Fig. 2 | Streamwater concentration changes before and after basalt application. **a**, Observed and estimated concentrations of alkalinity at W-2 (treatment watershed) and W-9 (reference watershed) from May 2022 through April 2025, respectively. The black dashed line separates the pre- and post-basalt application periods. Red and blue lines denote daily mean concentrations estimated by WRTDS, with shading representing 95% confidence intervals of the estimates; WRTDS models were calibrated with $n = 128$ discrete alkalinity samples at W-2, and $n = 330$ at W-9. **b**, Streamwater concentration changes after basalt application (orange circles), indicated by the differences between observed

concentrations and WRTDS-estimated concentrations for the no-basalt counterfactual. Blue lines with shaded areas represent locally estimated scatterplot smoothing (LOESS) mean fits⁷² with 95% confidence intervals. **c**, Streamwater C – q relations at W-2 before basalt application (May 2022 through June 2023; open circles) and during the first 6 months after basalt application (July 2023 through December 2023; solid squares). Data are coloured by month of year. Pre-application C – q data were fitted using a power-law model (grey dashed line, mean fit) with 95% confidence interval (grey shading); q (mm d^{-1}) represents the instantaneous discharge at the time of sample collection.

July through November 2023 (Fig. 2a). In the first 3 months following treatment, alkalinity and Ca^{2+} concentrations exceeded counterfactual estimates by an average of $267 \mu\text{eq l}^{-1}$ and $280 \mu\text{eq l}^{-1}$, respectively, peaking at $557 \mu\text{eq l}^{-1}$ and $696 \mu\text{eq l}^{-1}$ above the counterfactual in September (Fig. 2b), while measured Mg^{2+} and Na^{+} concentrations exhibited smaller increases ($<35 \mu\text{eq l}^{-1}$) relative to the counterfactual (Supplementary Fig. 1). The absence of similar responses in the W-9 reference watershed indicates that the observed concentration increases in W-2 stream were a direct result of basalt treatment (Fig. 2a

and Supplementary Fig. 1). The basalt treatment also affected streamwater C – q relations, weakening the pronounced pattern of dilution with increasing flow observed during the baseline (pre-basalt) period for both alkalinity and base cations (Fig. 2c and Supplementary Fig. 2). In addition to concentration increases, we observed small but statistically significant decreases in the elemental ratios of Ca/Na , $\text{HCO}_3^{-}/\text{Na}$ and Sr/Na (Supplementary Fig. 3), which are consistent with, but do not definitively demonstrate, a streamwater response to dissolution of the applied basalt²⁹.

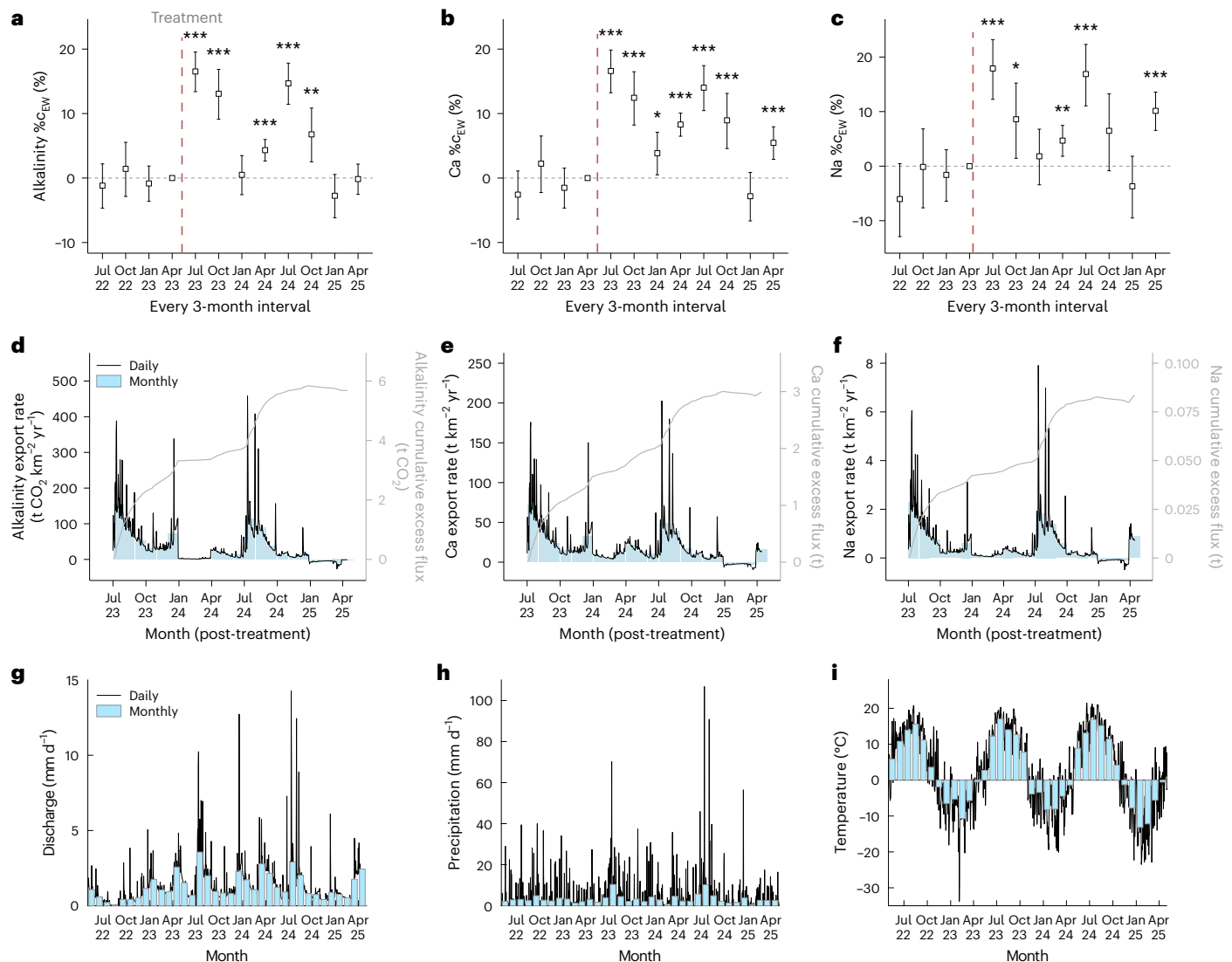


Fig. 3 | Temporal dynamics of the EW treatment effects, solute export rates and hydroclimatic variables. **a–c**, Percentage of streamwater concentration attributable to EW (%C_{EW}) for alkalinity (**a**), Ca (**b**) and Na (**c**), shown as mean treatment effects estimated from the GAM-DiD analysis. Error bars are 95% confidence intervals. Each GAM-DiD model was fit to $n = 842$ daily concentration values ($n = 434$ at W-2 and $n = 408$ at W-9). Significance was assessed with two-sided Wald tests on Watershed $\times$ Prepost interaction terms, without adjustment for multiple comparisons: *** $P \leq 0.001$, ** $P \leq 0.01$, * $P \leq 0.05$. Error bars without a symbol are not significant ($P > 0.05$). Exact P and n values for each

period are provided in Supplementary Tables 1 and 2. The vertical red dashed line separates the pre- and post-basalt application periods. **d–f**, Daily (solid black line) and monthly average (blue bar) export rates of the EW-derived alkalinity (**d**), Ca (**e**) and Na (**f**) estimated using equation (4) in the Methods. Grey solid lines represent the cumulative export flux of the basalt weathering products. **g–i**, Time series of daily (black lines) and monthly average (blue bars) discharge (**g**), precipitation (**h**) and temperature (**i**). In all panels, month labels (for example, 'Jul') correspond to the first day of that month.

The post-application enrichments in trace metal and silica concentrations provide further evidence for the rapid transport of basalt weathering products to the stream (Supplementary Fig. 2). Concentration increases of silica and lithium, which are both much more abundant in basalt than carbonates^{30,31}, mirrored the increases in alkalinity and base cations. Moreover, $\delta^7\text{Li}$ values declined from an average of $19.0 \pm 1.0\text{‰}$ during the pre-treatment period to $17.5 \pm 1.0\text{‰}$ within 3 months after treatment, and the pre- and post-treatment distributions were significantly different ($P < 0.001$; Supplementary Fig. 4), which is consistent with increased contributions of isotopically depleted Li derived from basalt-feedstock weathering (basalt $\delta^7\text{Li}$ value, $2.34 \pm 0.65\text{‰}$). Furthermore, concentrations of rubidium, which substitutes for potassium in feldspars and micas present in the basalt³², were similarly greater for equivalent stream discharges under post- relative to pre-treatment conditions. While iron, magnesium and strontium are released during dissolution of both basalt and carbonates³³, their concomitant increases

with silica, lithium and rubidium suggest freshly applied silicates as the most likely source of their increases in concentration immediately following the basalt application (Supplementary Fig. 2).

Preferential streamwater export of Ca relative to Na and Mg suggests either non-stoichiometric basalt dissolution and/or faster transport of Ca in response to the basalt treatment (Fig. 2 and Supplementary Fig. 2). The estimated increase in Ca concentration driven by EW is more than 20-fold greater than the corresponding increases in Na and Mg. Transport-related fractionation caused by cation-exchange reactions potentially contributes to the preferential Ca export. Competitive sorption³⁴ of cations on the Ca-dominated soil exchange complex at W-2 (mean Ca saturation $\sim 85\%$)³⁵ can lower the selectivity of the exchange sites for basalt-derived Ca and lead to the displacement of Ca by other basalt-derived base cations, particularly Mg, resulting in the earlier breakthrough of Ca relative to Mg. In addition, some Mg released from basalt may be retained in Mg-rich clays³⁶;

Table 1 | Export estimation of basalt weathering products with 95% confidence intervals

	Alkalinity as CO ₂ (t CO ₂ km ⁻² yr ⁻¹)	Ca (t km ⁻² yr ⁻¹)	Na (t km ⁻² yr ⁻¹)	Mg (t km ⁻² yr ⁻¹)	Si (t km ⁻² yr ⁻¹)
Jul–Sep 2023	102.55 (86.53, 117.30)	47.20 (39.27, 54.45)	1.50 (1.11, 1.84)	1.41 (1.09, 1.71)	2.83 (2.09, 3.53)
Oct–Dec 2023	46.18 (33.76, 57.46)	20.41 (14.21, 26.01)	0.39 (0.08, 0.66)	0.53 (0.26, 0.77)	0.87 (0.35, 1.36)
Jan–Mar 2024	2.87 (–12.27, 17.12)	8.98 (1.60, 15.86)	0.12 (–0.21, 0.41)	0.14 (–0.16, 0.42)	0.66 (0.11, 1.19)
Apr–Jun 2024	17.18 (10.97, 23.17)	15.21 (12.15, 18.14)	0.25 (0.11, 0.38)	0.37 (0.26, 0.49)	0.41 (0.17, 0.66)
Jul–Sep 2024	80.59 (65.00, 94.93)	35.05 (27.26, 42.18)	1.29 (0.90, 1.61)	1.19 (0.90, 1.45)	2.51 (1.82, 3.18)
Oct–Dec 2024	12.37 (4.55, 19.52)	7.88 (4.10, 11.30)	0.16 (–0.03, 0.32)	0.34 (0.20, 0.47)	0.58 (0.28, 0.86)
Jan–Mar 2025	–6.84 (–15.90, 1.66)	–2.71 (–6.93, 1.21)	–0.09 (–0.25, 0.06)	0.12 (–0.07, 0.31)	0.28 (–0.07, 0.61)
Apr 2025	–0.92 (–14.93, 12.44)	15.47 (8.55, 22.03)	0.88 (0.60, 1.13)	0.68 (0.43, 0.92)	1.06 (0.49, 1.62)
Annual mean	34.72 (30.39, 39.05)	18.71 (16.58, 20.83)	0.53 (0.44, 0.63)	0.59 (0.50, 0.68)	1.16 (0.97, 1.34)

Alkalinity derived from basalt weathering is expressed as the mass of CO₂ removed through EW (assuming 1 mol of CO₂ per 1 mol of HCO₃[–]).

however, this sink appears minor given the absence of ⁷Li enrichment in streamwater after treatment (Supplementary Fig. 4). Furthermore, basalt mineral dissolution is often non-stoichiometric. Clinopyroxene dissolution releases Ca preferentially over Mg (ref. 37), while plagioclase dissolution preferentially releases Ca over Na (ref. 38), a relation that is made more pronounced when considering the higher kinetic reactivity of Ca-rich plagioclase compared with its Na-rich counterparts, particularly under acidic conditions³⁹. Moreover, the applied basalt contained minor inorganic carbon (0.14 ± 0.02 wt%), which may contribute to preferential export of Ca to the stream. Together, competitive sorption on the Ca-dominant soil exchange complex, non-stoichiometric mineral dissolution and minor carbonate weathering may collectively enhance Ca export.

Seasonality of the EW treatment effect

The percentage of the observed streamwater concentrations attributable to EW (%c_{EW}) exhibited recurring seasonal patterns during the 2 years following basalt application, with stronger responses in summer and autumn and weaker responses in winter and spring (Fig. 3a–c). Results of the GAM-DiD modelling showed that the basalt treatment caused statistically significant changes in streamwater chemistry during July–September 2023, with EW accounting for 14% to 18% of alkalinity, Ca, Mg, Na and Si (%c_{EW}; Fig. 3a–c and Supplementary Fig. 5a). This strong treatment effect, coinciding with high discharge during summer storms (Fig. 3g,h), drove elevated EW-sourced export (that is, solute load per unit treated watershed area) in summer, which then declined as streamflow decreased later in the season (Fig. 3d–f). The EW treatment effect on concentration remained significant but weakened during October–December 2023 (Fig. 3a–c), and stream discharge also trended downwards during this period (Fig. 3g), leading to more than a 50% reduction in the export of alkalinity, base cations and Si (Fig. 3d–f and Table 1). During the wintertime (January–March 2024) when streamflow and temperature were at their lowest (Fig. 3g,i), EW treatment effects on solute concentrations were the weakest and statistically indistinguishable from zero, yielding small (in 2025, slightly negative) export estimates that reflect model uncertainty rather than a physical sink. EW-sourced alkalinity export sharply decreased from its summertime high of 102.55 t CO₂ km⁻² yr⁻¹ to less than 3 t CO₂ km⁻² yr⁻¹ (alkalinity expressed as removed CO₂, assuming 1 mol of CO₂ per 1 mol of HCO₃[–]), and base cations showed approximately proportional decreases (Fig. 3d–f and Table 1). The treatment effect on streamwater concentration increased again in April–June 2024 as temperatures rose and streamwater export was controlled by snowmelt-driven discharge (Fig. 3a–f). Comparable seasonal patterns in the EW-attributable concentration and export percentages were observed during the second year (July 2024–April 2025), though with slightly lower intensities (Fig. 3a–f and Table 1).

This recurring seasonal cycle of the EW treatment effect suggests that interactions among rainfall-runoff processes, watershed wetness

and temperature shaped the seasonal patterns in EW-derived solute concentrations and export. Comparison of C–q relations in the summer–autumn periods before and after basalt application reveals a weakening of the streamwater dilution pattern, which is consistent with a lowering of the difference in solute concentrations within slow-flow and quick-flow pathways that deliver water and solutes to the stream (Fig. 4a and Supplementary Fig. 6). By contrast, the C–q relations for winter–spring periods exhibited greater slopes and were similar before and after basalt application (Fig. 4a and Supplementary Fig. 6), which suggests comparatively large differences in solute concentrations between quick- and slow-flow pathways in the baseline period were preserved after the basalt treatment. Seasonal C–q relations fitted with a sigmoidal inverse power-law model (equation (8) in the Methods, with quick-flow/slow-flow endmembers) show that solute concentrations associated with slow-flow pathways remained stable year-round (alkalinity, 4,186 ± 228 µeq l⁻¹; Ca, 4,256 ± 455 µeq l⁻¹). However, concentrations of solutes transmitted via quick-flow pathways increased markedly during the summer–autumn seasons of the post-application period with alkalinity and Ca increasing by up to 1,200 µeq l⁻¹ (Fig. 4b and Supplementary Fig. 6).

In temperate catchments, such as W-2, quick flow comprises both new water (recent rainfall or snowmelt) that is delivered to the stream via overland and near-surface flow and resident (pre-event) soil water and shallow groundwater that is displaced primarily during hydrologic events and in diminishing quantities during recession to baseflow^{23,40}. EW-derived solutes in quick flow are likely associated with shallow pre-event water that has interacted extensively with the feedstock between storm periods. Lowland areas proximal to the stream with a shallow water table and high soil moisture are the probable sources of such solute-rich quick flow. By contrast, slow flow showed no detectable EW signals, indicating its upslope origin towards the watershed divide and correspondingly long residence times.

The summer–autumn increase in EW solute concentrations within quick-flow pathways implicates temperature as an important factor in regulating basalt-dissolution rates. Following the Arrhenius framework for silicate weathering applied by Brantley et al.²⁰ (equation (9), Methods), we find that the logarithm of the discharge-normalized EW-derived alkalinity export rate (E_{norm} ; equation (5), Methods) exhibits a linear relationship with the reciprocal of temperature, yielding an apparent activation energy, E_{app} , of 58.5 ± 15.5 kJ mol⁻¹ ($R^2 = 0.47$, $P = 0.002$; Fig. 5), which represents the temperature sensitivity of basalt dissolution. This E_{app} value is within the expected range for basalt weathering (40–90 kJ mol⁻¹) (ref. 20) and reveals that dissolution of the basalt feedstock responds strongly to the seasonal temperature variation observed at W-2. This E_{app} value exceeds laboratory dissolution activation energies for Ca-bearing plagioclases⁴¹, the primary alkalinity-producing minerals in basalt, indicating enhanced temperature sensitivity at the watershed scale. This is consistent with the pattern identified by Brantley et al.²⁰, in which field E_{app} values systematically exceed laboratory values as

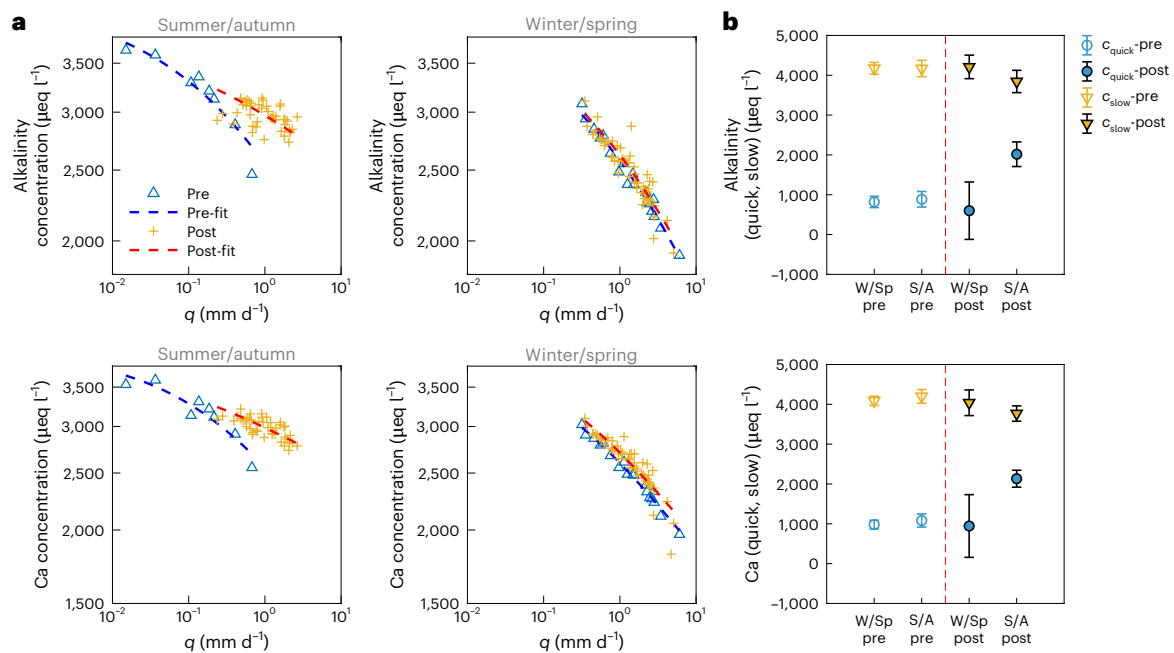


Fig. 4 | Seasonal C–q relations and estimated concentrations of slow-flow/quick-flow endmembers pre- and post-basalt application. a, Observed C–q relations for alkalinity and Ca for summer/autumn and winter/spring periods before (blue triangles) and after basalt application (orange crosses), with corresponding model mean fits (dashed lines). **b**, Mean alkalinity and Ca concentrations of quick-flow (blue) and slow-flow (orange) endmembers for winter/spring and summer/autumn periods before (open symbols) and

after basalt application (filled symbols), separated by the vertical red dashed line, estimated from inversion of equations (6)–(8) in the Methods; error bars represent 95% confidence intervals of the endmember estimates, computed from the residuals and Jacobian of the least-squares fit (nlparci, MATLAB R2023b). $n = 19, 8, 55$ and 49 for pre-application winter/spring, pre-application summer/autumn, post-application winter/spring and post-application summer/autumn, respectively.

additional temperature-dependent processes couple to dissolution, such as seasonal variation in soil $p\text{CO}_2$ levels. Neither temperature nor E_{norm} was correlated with discharge at the monthly scale (Supplementary Fig. 7), indicating that the inferred temperature dependence of EW-derived alkalinity is not an artefact of hydrologic variability.

Export rates of basalt weathering products and associated CDR

On an annual basis, EW-derived alkalinity export averaged $9.47 \text{ t C km}^{-2} \text{ yr}^{-1}$, equivalent to a maximum CDR export rate of $34.72 \text{ t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$ (assuming 1 mol of CO_2 per 1 mol of HCO_3^- and attributing all of it to silicate weathering; Table 1), which is among the highest rates reported in studies that quantify CDR via changes in alkalinity or cation fluxes in streams, leachate waters or pore waters^{36,42–46}. Previous watershed-scale studies reported of $2.5\text{--}13 \text{ t CO}_2 \text{ km}^{-2}$ over 15 years after wollastonite addition (3.44 t ha^{-1}) in a temperate forest⁴², while another study reported an average of $10 \text{ t CO}_2 \text{ km}^{-2}$ across three small watersheds ($<0.02 \text{ km}^2$) within a tropical oil palm plantation after three annual applications of crushed andesite (50 t ha^{-1}) (ref. 43). In addition, the annual CDR rate in our field trial is approximately equal to the median CO_2 consumption rate ($33 \text{ t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$) estimated from bicarbonate measurements made in rivers draining 12 basaltic watersheds distributed across volcanic arcs, oceanic islands and continental flood basalt provinces⁴⁷. Lysimeter and mesocosm studies usually yield lower rates ($1\text{--}26 \text{ t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$), even though many used higher rock application rates of up to 200 t ha^{-1} (refs. 11,44,45,48,49). The lack of leachate EW signals in some of these experimental studies likely represents a combination of factors, including short experimental durations, the absence of natural structure and associated transport pathways in repacked soils, and, likely foremost, the use of acidic soils that will at least temporarily scavenge base cations and neutralize alkalinity. Recent field-based studies that estimated CDR via soil-based techniques report higher removal ($10\text{--}370 \text{ t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$) (refs. 50–53); however, these studies used higher

feedstock application rates and do not account for processes, such as ion exchange and carbonate precipitation, that may reversibly or irreversibly reduce CDR efficiency.

Over 22 months, the EW-derived alkalinity export was $63.8 \text{ t CO}_2 \text{ km}^{-2}$ (5.7 t over the 8.9 ha treated area; Fig. 3d), representing approximately 11% of the maximum CDR potential of the applied basalt ($582 \text{ t CO}_2 \text{ km}^{-2}$, 51.8 t CO_2 over the 8.9 ha treated area; equation (1) in the Methods). While the applied basalt contained minor inorganic carbon ($0.14 \pm 0.02\%$, or 0.22 t in total), its maximum dissolution would account for 28% of the exported alkalinity, of which half (14%) derives from the carbonate mineral itself, and the other half (14%) derives from CO_2 consumed during carbonate weathering. This indicates that most of the alkalinity export was generated through silicate weathering. In total, this corresponds conservatively to 9.5% of the basalt's CDR potential, with 8% from silicate weathering and 1.5% from CO_2 consumed during the potential carbonate weathering. This value approaches the upper limit of realizable export as predicted by soil-column, reactive-transport models that account for cation sorption and secondary mineral formation. Under such constraints, alkalinity export is generally estimated to be $<10\%$ after 2 years in most areas of the United States⁵⁴. One reason for the early emerging signals of weathering in this study, despite the high pH and cold climate, is the high initial base saturation of soils in W-2 (85% Ca), indicating that the potential maximum loss to cation sorption may be lower compared with US averages; suitable hydrologic conditions may have also contributed to the early emerging signals of weathering. Collectively, these findings demonstrate that weathering products can be exported efficiently from headwater catchments where transport to streams and rivers occurs over seasonal-to-annual timescales.

Implications for EW applications at watershed scale

This study demonstrates that, despite the slow weathering rates expected in a northern climate, watersheds can be natural integrators of EW signals and provide a scalable unit for monitoring, reporting

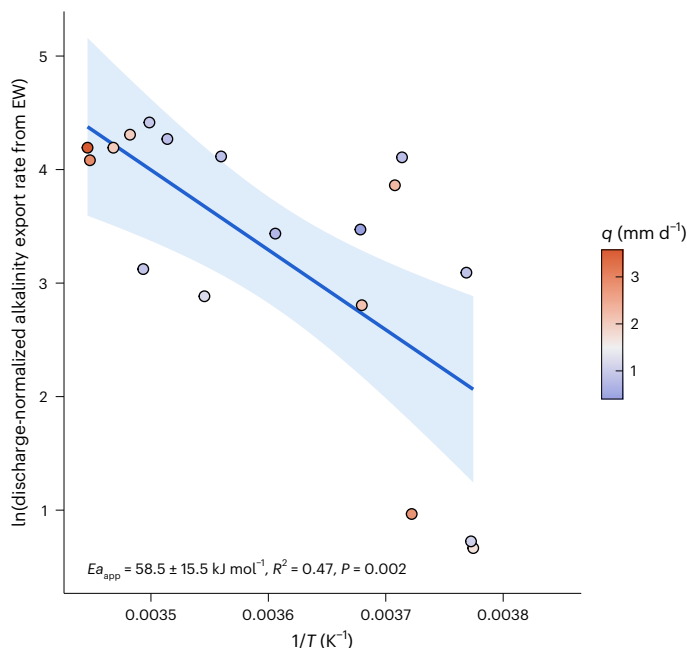


Fig. 5 | Arrhenius relationship between discharge-normalized EW-derived alkalinity export rate and temperature at W-2 watershed. Discharge-normalized EW-derived alkalinity export rate (E_{norm} , $\text{t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$; equation (5), Methods) is plotted against temperature (T , K). Each point represents a monthly average, coloured by stream discharge q (mm d^{-1}). The apparent activation energy ($E_{a,app} = 58.5 \pm 15.5 \text{ kJ mol}^{-1}$) was estimated as the product of $-R$ (gas constant, $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$) and the slope of the linear regression of $\ln(E_{norm})$ against $1/T$ ($R^2 = 0.47$, $P = 0.002$). The shaded area indicates the 95% confidence interval of the regression.

and verification by capturing the combined effects of basalt weathering rate, hydrologic transport and climatic variability (for example, temperature and precipitation) on the export of EW-derived alkalinity and base cations. The EW treatment effects estimated by the watershed approach (with GAM-DiD) showed distinct, recurring seasonal patterns, with broadly comparable intensities in both years when comparing the same seasons, despite a slight overall decrease in the second year. This temporal consistency indicates that comparable hydrologic and environmental conditions in 2023 and 2024 (Fig. 3g–i) exerted a stronger control on the weathering signal than processes intrinsic to the weathering material itself, such as the depletion of reactive surface area or secondary phase formation, at the timescale of about 2 years^{55,56}.

The EW signal in W-2 streamwater was detected within weeks of basalt application, consistent with rapid delivery of basalt weathering products along quick-flow paths originating close to the stream, yet only 9.5–11% of the potential maximum CDR was exported as alkalinity over 2 years. The mean residence time of water within W-2 catchment was 1.3 years⁵⁷. Our water-isotope analysis (Methods) showed that $23 \pm 2\%$ of streamflow comprised water younger than 2.3 ± 0.8 months, implying that most water resides in longer-term catchment storage. Thus, it is likely that EW products transported along slow-flow pathways originating upslope and farther from the stream had yet to emerge and may be difficult to resolve in the future because of dilution. These results imply that silicate deployments that preferentially target near-stream corridors may be most responsive to streamwater-based monitoring, reporting and verification, while also suggesting that extended monitoring windows will likely be necessary to observe weathering-signal tails and avoid underestimating CDR.

Near-stream corridors often have perennially moist soils that favour rapid weathering and may be ideal locations for repeated silicate applications if potential reductions in CDR efficiency associated with high-silicate loadings are well understood. At W-2, for example,

the calcite saturation index of streamwater exceeds 0 at an alkalinity of $-2,000 \mu\text{eq l}^{-1}$ and increases by approximately 0.1–0.2 units per $500 \mu\text{eq l}^{-1}$ once alkalinity exceeds $-2,000 \mu\text{eq l}^{-1}$ (Supplementary Fig. 8). Repeated annual applications could therefore push W-2 waters towards seasonal (that is, summertime) calcite precipitation, reducing net alkalinity export and effective CDR. Catchments characterized by conditions conducive to carbonate precipitation or secondary-mineral formation may require less frequent applications or reduced doses. However, the presence of soil calcite within the W-2 catchment makes it a unique environment relative to many targets for EW.

Our results demonstrate temperature sensitivity of alkalinity export and hence CDR. These findings indicate that CDR efficiency will be comparatively low in temperate watersheds, even where year-round precipitation promotes runoff and weathering-product transport. Accordingly, monitoring, reporting and verification frameworks that rely on streamwater chemistry should account for seasonal variability to avoid misestimation of EW-derived alkalinity export. While this work begins to clarify how interactions between temperature and hydrologic variation affect EW solute deliveries to surface waters and associated CDR, additional watershed-scale trials across diverse hydroclimatic settings are needed. Integrating such observations with an emerging set of watershed-scale reactive transport models that accurately and consistently represent watershed-runoff processes and geochemical reactions would be an important next step in advancing a framework for informing decisions on optimal locales for EW deployments.

Methods

Site description

The study site, the W-2 watershed, is a 59-ha headwater catchment situated within the Sleepers River Research Watershed in Danville, Vermont, USA. It is a low-elevation (285 m to 377 m) agricultural catchment (average slope -7.3°), with 27% forest and 73% hayfield and pasture²³. The area experiences a cold, humid continental climate, with mean annual precipitation of $1,050 \text{ mm yr}^{-1}$ and mean annual air temperature of 5.7°C . The mean daily discharge from the W-2 stream averages $0.76 \pm 0.19 \text{ mm d}^{-1}$, with approximately two-thirds of annual discharge occurring as baseflow²⁹. While discharge peaks in spring during snowmelt, heavy storms during summer and autumn may drive discharge to levels more than an order of magnitude above baseflow. The stream weir location for discharge measurement, where water samples were also collected, is Lat: 44.45958 Lon: -72.0920 .

The watershed is underlain by fine-grained, silty, calcareous till deposited during the Wisconsin glacialiation⁵⁸. Beneath the till is the Late Silurian to Early Devonian Waits River Formation, which consists of quartz-mica schist with beds of quartz-rich limestone⁵⁹. Till thickness ranges from less than 2 m near the hilltops to over 10 m downslope towards the stream. The soil in the upland hayfield is a sandy loam, whereas the downslope pasture soil is a loam with approximately equal sand and silt contents, with pH ranging from 4.9 to 7.7 (median pH 6.1) (ref. 35).

Basalt application and characteristics

We applied Pioneer Valley ‘BrixBlend’ Basalt powder at a rate of 20 t ha^{-1} (without tilling) to an 8.9-ha area of hayfield and pasture in the southern portion of the W-2 watershed (Fig. 1) in June 2023. The basalt feedstock was sourced from the Holyoke Basalt of Newark Supergroup^{60,61} (MA, CT, USA; supplied by Rock Dust Local, Bridport, Vermont, USA). The moisture content of basalt powder was $10.1 \pm 0.2\%$ at the time of application. Modal mineralogy (thin-section point counts) of the basalt was 35.1% clinopyroxene, 33.7% plagioclase, 10.6% sericite, 9.2% chlorite, 6.7% actinolite, 3.8% opaque minerals and 0.9% quartz. The basalt feedstock consisted primarily of SiO_2 (51.6%), Al_2O_3 (13.6%) and Fe_2O_3 (13.2%), with notable contents of CaO (9.2%), MgO (5.8%) and Na_2O (2.98%) (ref. 62) (Supplementary Table 3). On the basis of this oxide content, the estimated stoichiometry of the basalt is $\text{Ca}_{0.36}\text{Mg}_{0.31}\text{Na}$

$0.209\text{Fe}_{0.36}\text{Si}_{1.87}\text{Al}_{0.58}\text{O}_6$ (molar mass of 211 g mol^{-1}). The total inorganic carbon content was $0.14 \pm 0.02\%$, measured on the Eltra CS Analyzer. The p80 value (80% of the particles have a diameter less than or equal to) of the basalt powder equalled $297\text{ }\mu\text{m}$. The specific surface area of the basalt was $4.3\text{ m}^2\text{ g}^{-1}$, measured using the Brunauer–Emmett–Teller N_2 -adsorption method on an Anton Paar (Graz, Austria) Nova 800.

The CDR potential ($\text{t CO}_2\text{ km}^{-2}$) of the basalt was calculated by using the following equation:

$$\text{CDR}_{\text{potential}} = \sum_{i=\text{Ca, Mg, Na}} n_i \cdot \frac{M_{\text{CO}_2}}{M_{\text{basalt}}} \cdot R_{\text{app}} \cdot (1 - \theta) \quad (1)$$

where n_i (–) is the mole of charge of each major cation (that is, Ca, Mg and Na) per mole of basalt ($\text{Ca}_{0.36}\text{Mg}_{0.31}\text{Na}_{0.209}\text{Fe}_{0.36}\text{Si}_{1.87}\text{Al}_{0.58}\text{O}_6$); M_{CO_2} (g mol^{-1}) and M_{basalt} (g mol^{-1}) are the molar mass of CO_2 and basalt, respectively; θ (%) is the moisture content of basalt; and R_{app} (t km^{-2}) is the basalt application rate.

Streamwater chemistry and meteorological observations

Streamwater samples were collected biweekly or monthly before basalt application from May 2022 through June 2023, and weekly or twice per week after basalt application from July 2023 through April 2025, with more intensive sampling during storms (hourly). The collected streamwater samples were either filtered on-site immediately or filtered in the laboratory within 24 h through $0.22\text{ }\mu\text{m}$ syringe filters, and divided into subsamples in acid-washed, filtered-streamwater-rinsed, high-density polyethylene bottles for major cations (calcium (Ca^{2+}), magnesium (Mg^{2+}), sodium (Na^+), potassium (K^+)), alkalinity and trace metal analysis. The samples for trace metal analysis were acidified by adding 50% nitric acid. All the samples were stored at $4\text{ }^\circ\text{C}$ until analysis. All measurements were performed at the Yale Analytical and Stable Isotope Center (YASIC) following the methods described in ref. 29. Aliquots from selected samples were further processed for Li separation and measurements at the Yale Geochemistry Center. The samples were first digested with aqua regia, evaporated and redissolved in 1 ml of 0.2 N HCl, and then Li was separated following the methods described in ref. 63. The Li isotopic composition measurements were performed on a Thermo Finnigan Neptune Plus ICP-MS and the Li isotopic compositions are reported relative to the L-SVEC-1 lithium carbonate standard, NIST 8545. A typical standard error of a single measurement was 0.07% (1s.d.) and the external precision was better than 0.08% (1s.d.).

In addition to our own observations, we used published measurements of stream chemistry made from March 1992 through May 2017 at the W-2 watershed and measurements of stream discharge and chemistry from January 2019 through April 2025 at the W-9 watershed published in a U.S. Geological Survey (USGS) data release⁶⁴. Watershed W-9 is a 41-ha forested headwater catchment located 6 km from W-2 in the Sleepers River Research Watershed. Daily estimates of air temperature and precipitation for the W-2 and W-9 watersheds were extracted from the Parameter-elevation Regressions on Independent Slopes Model (PRISM) database⁶⁵.

Estimating streamwater chemistry for a no-basalt counterfactual with WRTDS

We applied WRTDS to estimate streamwater chemistry at the W-2 watershed for a counterfactual scenario representing conditions without the basalt application. The model was trained on long-term data from March 1992 through June 2023, and the data from February 2020 through May 2022 were used as a validation dataset to compare simulated and observed values (Supplementary Fig. 9). The calibrated model was used with W-2 daily discharge measurements from July 2023 through April 2025 as one means to estimate how streamwater solute

concentrations at W-2 would vary during the post-treatment period in the absence of the basalt application. The WRTDS method was implemented using the EGRET R package^{26,66}. Briefly, the daily streamwater solute concentrations were simulated in WRTDS as

$$\ln(c) = \beta_0 + \beta_1 \ln(q) + \beta_2 t + \beta_3 \sin(2\pi t) + \beta_4 \cos(2\pi t) + \varepsilon \quad (2)$$

where c is the estimated concentration ($\mu\text{eq l}^{-1}$), β_i are the regression coefficients, q is the daily mean discharge (mm d^{-1}), t is time in decimal years, and ε is the unexplained variation. This method uses a weighted regression, where the relevance of each observation to the estimation point is defined by the distance in time, season and discharge between the observation and the estimation point, thereby generating a unique set of parameters for every combination of q and t values^{26,27}. The WRTDS model configuration and fit diagnostics for each solute are provided in Supplementary Table 4.

Estimation of basalt treatment effect using a difference-in-differences approach

To quantify basalt-treatment effects on streamwater chemistry, we implemented a GAM-DiD approach that leverages stream chemistry observations from a reference watershed to control for shared hydroclimatic variability and isolate concentration changes attributable to EW. Watershed W-9 was chosen as the reference because of its similar climate and parallel hydrologic and hydrochemical responses relative to W-2 (Fig. 1). The model structure is as follows:

$$\begin{aligned} \ln c_i = & \beta_0 + s_1(\ln q_i) + s_2(\text{PPT}_i) + s_3(T_i) + s_4(\text{Month}_i, \text{by} = \text{Watershed}_i) \\ & + \beta_1 \cdot \text{Watershed}_i + \beta_2 \cdot \text{Prepost}_i + \beta_3 \cdot (\text{Watershed}_i \times \text{Prepost}_i) + \varepsilon_i \end{aligned} \quad (3)$$

where $\ln c_i$ is the response variable (natural-log-transformed solute concentration c_i ($\mu\text{eq l}^{-1}$) for sample or daily estimate i from either the treatment or the reference watershed; s_1, s_2, s_3 and s_4 are the spline-based smooth functions used to model non-linear relationships between each predictor and $\ln c_i$; $\ln q_i$ is the natural-log-transformed daily mean discharge q_i (mm d^{-1}); PPT_i (mm d^{-1}) is the precipitation; T_i ($^\circ\text{C}$) is the daily average air temperature; $(\text{Month}_i, \text{by} = \text{Watershed}_i)$ indicates the month effect by watershed (treatment or reference); Watershed_i is the binary indicator for treatment and reference watershed (0 = reference, 1 = treatment); Prepost is a categorical indicator for pre- and post-treatment period, where $-4, -3, -2$ and -1 (reference level) correspond to the pre-treatment period (3-month intervals from July 2022 to June 2023), and $0-7$ represent post-treatment periods (3-month intervals, except for April 2025, which covers 1 month owing to the absence of W-9 water chemistry records thereafter); $\text{Watershed} \times \text{Prepost}$ is the interaction term representing the core difference-in-differences (DiD) contrast that captures the difference in differences (post versus pre) between treatment and reference watersheds. Fixed effects comprise an intercept β_0 (expected log concentration in the baseline pre-treatment period for the reference watershed at reference covariate levels), a watershed indicator β_1 (time-invariant treatment-reference difference at baseline), period ('Prepost') indicators β_2 (shifts common to both watersheds for each period relative to baseline), and the $\text{Watershed} \times \text{Prepost}$ interactions β_3 , which quantify the DiD for each period. The variable ε_i is the residual error.

The coefficients in equation (3) are on the natural-log scale, such that exponentiation yields multiplicative effects on concentration. Of particular importance is the quantity $(\exp(\beta_3 - 1)) / (\exp(\beta_3)) \times 100$, which gives the percentage of streamwater concentration attributable to EW; we denote this as $\%c_{\text{EW}}$ (%). The 95% confidence intervals of $\%c_{\text{EW}}$ were obtained by applying the same transformation to $\beta_3 \pm 1.96 \times \text{SE}$. This estimated treatment effect and the predicted post-treatment concentration (equation (3), with $\text{Watershed} = 1$ and $\text{Prepost} = 0-7$)

from the GAM-DiD model were used to calculate the export rate of the basalt weathering products via stream runoff (equation (4)).

The statistical significance of each parametric coefficient was assessed with two-sided Wald tests as implemented in *mgcv*, without adjustment for multiple comparisons. The parametric coefficients of the GAM-DiD models, including the intercept, watershed indicator, period indicators, sample sizes and DiD interaction terms, are reported in Supplementary Tables 1 and 2. The summary of smooth terms, including estimated degrees of freedom, *F*-statistics and *P* values, is provided in the deposited dataset⁶⁷. Partial effect plots for the continuous smooth terms (discharge, precipitation and temperature) across all five solutes are shown in Supplementary Fig. 10. The GAM-DiD model was implemented using the *mgcv* package⁶⁸ (version 1.9.1) in R.

To enable a DiD analysis despite the mismatched sampling dates of the two watersheds, we used WRTDS daily estimates for both watersheds during the pre-treatment period and for the reference watershed (W-9) during the post-treatment period. The W-9 WRTDS model was trained using W-9 data from January 2019 through June 2023, with data from July 2023 through April 2025 reserved as a validation dataset. The simulated and observed values showed good agreement during this validation period (Fig. 2 and Supplementary Fig. 9). The daily estimates that corresponded to times of measurements at W-2 (where available) were used as reference-watershed concentrations in the post-treatment DiD analysis.

Estimation of the export rates of basalt weathering products and associated CDR

We calculated the export rates E_i ($\text{t km}^{-2} \text{yr}^{-1}$) of basalt weathering product i ($i = \text{Ca, Na, Mg and Si}$) and associated CDR ($i = \text{CO}_2$, with alkalinity expressed as removed CO_2 , assuming 1 mol of CO_2 per 1 mol of bicarbonate) by integrating the product of the predicted post-treatment concentration $c_{\text{post},i}$ ($\mu\text{eq l}^{-1}$) (c_i in equation (3), with $\text{Watershed} = 1$ and $\text{Prepost} = 0-7$) and the percentage of streamwater concentration attributable to EW of solute i predicted from the GAM-DiD model ($\%c_{\text{EW},i}$, equation (3)), and daily mean discharge q (mm d^{-1}) (normalized to the whole catchment area $A_{\text{catchment}}$ (km^2), 0.59 km^2) over the post-treatment period from t_0 to t . The integrated flux was then normalized to the basalt treatment area A_{basalt} (km^2 , 0.089 km^2) and the post-treatment period ($t - t_0$) (days), such that

$$E_i = \frac{1}{t - t_0} \int_{t_0}^t c_{\text{post},i} \cdot \frac{\%c_{\text{EW},i}}{100} \cdot q \cdot A_{\text{catchment}} \cdot \frac{1}{A_{\text{basalt}}} \cdot \frac{M_i}{n_i \times 10^6} \cdot 365 \, dt \quad (4)$$

where M_i (g mol^{-1}) and n_i (–) are the molar mass and charge number of solute i , respectively, and 10^6 in the denominator represents the conversions of microequivalents to equivalents. For the calculation of the EW-derived alkalinity export rate in the form of removed CO_2 , $n_i = 1$ (assuming 1 mol of CO_2 per 1 mol of bicarbonate), and $M_i = 44 \text{ g mol}^{-1}$. The post-treatment period ($t - t_0$) for the estimation of the average export rate spans from July 2023 through April 2025, and the sub-period estimates were calculated either monthly (Fig. 3d–f) or at 3-month intervals (Table 1; except for the final sub-period, which ended in April 2025 and was therefore shorter than 3 months).

The monthly discharge-normalized export rate E_{norm} ($\text{t km}^{-2} \text{yr}^{-1}$) is defined as the monthly EW-derived alkalinity flux from the GAM-DiD analysis (E_i , equation (4)), divided by the corresponding monthly average stream discharge $q_{\text{monthly},i}$ (mm d^{-1}) and then multiplied by the 22-month post-treatment average discharge q_{ave} (mm d^{-1} , 1.5 mm d^{-1}). This normalization removes month-to-month hydrologic variability while preserving the long-term flux magnitude²⁰, so that any residual seasonal signal in E_{norm} represents temperature-dependent weathering kinetics rather than variation in discharge (equation (5)).

$$E_{\text{norm}} = \frac{E_i}{q_{\text{monthly},i}} \cdot q_{\text{ave}} \quad (5)$$

Endmember mixing analysis

To trace basalt-derived solutes and quantify flow-path contributions to the W-2 stream, we developed a quick-flow/slow-flow mixing model with a sigmoidal inverse power-law function (equations (6)–(8)). Here we define two flow-path endmembers of streamwater composition: one is slow-flow water composed primarily of deeper groundwaters with longer residence times and the other is quick-flow water composed of near-surface and overland-flow waters with shorter residence times. Quick-flow/slow-flow separation has been used with analysis of stream C– q relations to infer flow paths by which water is transported through the watershed^{69,70}. A non-linear relation between stream discharge q and the slow- and quick-flow fractions (that is, f_{slow} and f_{quick}) can be derived under the assumption that f_{slow} decreases with q and asymptotically approaches zero as q approaches infinity and that the sum of f_{slow} and f_{quick} always equals unity (equation (6), Supplementary Fig. 11a). We combined this sigmoidal inverse power-law function with an endmember mixing equation (equation (7)) to estimate a relation between stream discharge and stream concentration (equation (8)), such that

$$f_{\text{slow}} = \frac{1}{1 + a_{\text{slow}} \cdot q^{b_{\text{slow}}}} \quad (6)$$

$$c_{\text{stream}} = c_{\text{slow}} \cdot f_{\text{slow}} + c_{\text{quick}} \cdot (1 - f_{\text{slow}}) \quad (7)$$

$$c_{\text{stream}} = (c_{\text{slow}} - c_{\text{quick}}) \cdot \left(\frac{1}{1 + a_{\text{slow}} \cdot q^{b_{\text{slow}}}} \right) + c_{\text{quick}} \quad (8)$$

where f_{slow} (–) is the fractional contribution of slow-flow water to streamflow, q (mm d^{-1}) is discharge of the stream, a_{slow} (–) and b_{slow} (–) are fitted values to identify the f_{slow} change with q in the watershed. c_{stream} ($\mu\text{eq l}^{-1}$) is the streamwater concentration, c_{slow} ($\mu\text{eq l}^{-1}$) is the slow-flow endmember concentration and c_{quick} ($\mu\text{eq l}^{-1}$) is the quick-flow endmember concentration.

By fitting equations (6)–(8) to streamwater chemistry data for the pre-treatment period (January 2008–May 2017 and May 2022–June 2023), we estimated the endmember concentrations of c_{quick} and c_{slow} of the solutes (that is, Ca, Mg, Na and alkalinity) together with the parameters a_{slow} and b_{slow} (Supplementary Fig. 6). With the estimated values of a_{slow} and b_{slow} from the pre-treatment data, we then estimated c_{quick} and c_{slow} of each solute after basalt application. This analysis provides a way to explore whether basalt weathering signals (for example, elevated alkalinity and Ca concentration) in the stream are more closely associated with quick-flow or slow-flow pathways, and how such signals may evolve after basalt treatment. While we do not assume that the relation between flow pathways and discharge is precise or universally transferable across catchments or time periods, this approach can be used to help identify the potential origins of observed streamwater concentration changes. Pre-treatment non-linear parameters a_{slow} and b_{slow} were estimated by non-linear fitting (*fmincon*). Post-treatment endmember concentrations for each season-period group were then estimated with these parameters fixed using constrained least squares (*lsqcurvefit*), with 95% confidence intervals computed from the fit residuals and Jacobian (*nplarci*), in MATLAB R2023b.

Estimation of apparent activation energy of basalt weathering

The apparent activation energy E_{app} (J mol^{-1}) of basalt weathering was estimated from the temperature dependence of EW-derived alkalinity export using the Arrhenius relationship:

$$\Omega = \Omega_{\text{ref}} \cdot \exp\left(-\frac{E_{\text{app}}}{RT}\right) \quad (9)$$

where Ω ($\text{t km}^{-2} \text{yr}^{-1}$) is the basalt weathering rate, Ω_{ref} is the weathering rate at a reference temperature T_{ref} (K), R is the gas constant ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$), and T (K) is the mean air temperature. We used the monthly discharge-normalized EW-derived alkalinity export rate E_{norm}

($\text{t km}^{-2} \text{ yr}^{-1}$) (refer to equations (4) and (5) above) as a proxy for Ω , with average air temperature for each interval. Alkalinity was used rather than individual cation fluxes (for example, Ca^{2+} , Mg^{2+} and Na^+) because alkalinity directly represents the total basalt weathering rate by integrating the charge-balanced contribution of all dissolving mineral phases and by avoiding biases from non-stoichiometric dissolution and from differential cation retention through exchange and biological uptake in the riparian soils. Ea_{app} was estimated as the product of $-R$ and the slope of the linear regression of $\ln(\Omega)$ against $1/T$. Temperature and discharge were not correlated at the monthly scale ($R^2 = 0.1$, $P = 0.16$, $n = 22$), indicating that this estimate is not strongly biased by discharge–temperature covariation.

Quantifying the young water fraction of streamflow

We calculated the young water fraction (F_{yw}) of the W-2 streamflow using $\delta^{18}\text{O}$ isotope values from precipitation and streamwater from March 1992 through June 2011⁶⁴, following a published approach⁷¹. In the following equation, F_{yw} is defined as the proportion of the transit-time distribution younger than a threshold age and can be estimated from the ratio of the amplitudes of tracer signals (for example, $\delta^{18}\text{O}$ isotope) in streamwater and precipitation for gamma functions with shape parameter α between 0.2 and 2 (ref. 71):

$$F_{yw} = A_s/A_p \quad (10)$$

where A_s (–) and A_p (–) are the amplitudes of the $\delta^{18}\text{O}$ signals in streamwater and precipitation, respectively.

The amplitude and phase of the seasonal $\delta^{18}\text{O}$ from precipitation and streamwater were estimated by non-linear fitting of

$$c(t) = A \sin(2\pi ft - \varphi) + k \quad (11)$$

where A (–) is the amplitude, f (yr^{-1}) is the frequency, t (yr) is time, φ (–) is the phase shift and k (–) is the vertical shift.

The calculated gamma shape parameter α for the W-2 transit-time distribution is 0.81, estimated by solving equation (11) using Newton's method:

$$\varphi_s - \varphi_p = \alpha \arctan \sqrt{\left(\frac{A_s}{A_p}\right)^{-\frac{2}{\alpha}} - 1} \quad (12)$$

where φ_s (–) and φ_p (–) are the phase shifts for streamwater and precipitation, respectively. With the calculated α at W-2 falling within the range of 0.2–2, we reported F_{yw} of the W-2 streamflow as the fraction of water younger than 2.3 ± 0.8 months⁷¹. Gaussian error propagation was used to estimate uncertainties.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The W-2 streamwater discharge, major-ion chemistry, lithium isotope data and the fitted GAM-DiD smooth terms are available via Zenodo at <https://doi.org/10.5281/zenodo.20752184> (ref. 67).

Code availability

The code for the GAM-DiD analysis is available via GitHub at <https://github.com/sfcmarie/GAM-DiD>.

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Author contributions

F.S. and J.E.S. conceived, designed and established the experiment. J.E.S., P.A.R. and N.J.P. supervised the project. F.S. led the field sampling, conducted the water chemistry analysis with R.A.R., W.A.M.-B., A.A.M. and E.G., and performed data analysis and modelling. B.K.-A. performed lithium isotope analysis. W.T. performed young water fraction quantification. J.B.S. provided the long-term W-9 streamwater chemistry data. F.S. drafted the paper, and R.A.R., T.J.S., W.T., B.K.-A., Q.Z., J.B.S., P.A.R., N.J.P. and J.E.S. contributed to editing and revision. All authors approved the final version.

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Competing interests

The authors declare no competing interests. N.J.P. was a co-founder of the CDR company Lithos Carbon, but has no financial ties to the company.

Additional information

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Data analysis We applied WRTDS to estimate streamwater chemistry at the W-2 and W-9 watersheds, implemented using the EGRET R package. To quantify basalt-treatment effects on streamwater chemistry, we implemented a Generalized Additive Model coupled with a Difference-in-Differences (GAM-DiD) approach that leverages stream chemistry observations from a reference watershed. The GAM-DiD model was implemented using the mgcv package in R. The code is available at <https://github.com/sfcmarie/GAM-DiD>. R version 4.3.3 was used for all the analysis.
The endmember mixing analysis was performed in MATLAB R2023b, using the fmincon function for pre-treatment nonlinear parameter estimation and lsqcurvefit function for post-treatment endmember estimation.
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We used published measurements of stream chemistry made from March 1992 to May 2017 at W-2 watershed (<https://www.usgs.gov/data/aqueous-chemistry-database-sleepers-river-research-watershed-danville-vermont-1991-2018>), as well as measurements of stream discharge and chemistry made from January 2019 to April 2025 (which will be published soon) at W-9 watershed from the U.S. Geological Survey (USGS) aqueous chemistry database of the Sleepers River Research Watershed, Danville, Vermont. The W-2 streamwater discharge, major-ion chemistry, lithium isotope data, and the fitted GAM-DiD smooth terms are available in Zenodo at <https://doi.org/10.5281/zenodo.20752184>. Daily estimates of air temperature and precipitation for the W-2 and W-9 watersheds were extracted from the Parameter-elevation Regressions on Independent Slopes Model (PRISM) database (<https://prism.oregonstate.edu/>).

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Study description

We conducted a watershed-scale enhanced rock weathering (ERW) experiment at W-2 watershed, a headwater catchment located in the Sleepers River Research Watershed, VT. Following a one-year baseline monitoring period (May 2022 to Jun 2023), we applied Pioneer Valley Basalt powder (hereafter "basalt") in June 2023 over 8.9 ha of the southern portion of the catchment (15% of the total catchment area) at a rate of 20 t ha⁻¹. We isolated the basalt-weathering signal from background water chemistry by analyzing streamwater concentration–discharge (C–Q) relations before and after basalt application and by comparing streamwater observations to a no-basalt counterfactual estimated from WRTDS (Weighted Regressions on Time, Discharge, and Season). We then employed a Generalized Additive Model coupled with a Difference-in-Differences (GAM-DiD) approach that leveraged observations from a nearby reference watershed (W-9) to quantify seasonal dynamics in the streamwater export of basalt-derived alkalinity, base cations and associated CDR rates. Additionally, we applied a novel endmember mixing analysis to evaluate the effects of basalt

	treatment on seasonal shifts in weathering-product concentrations within quick-flow and slow-flow pathways that transmit solutes to the stream.
Research sample	Streamwater samples were collected to monitor the effects of enhanced rock weathering (ERW) in the watershed. Water samples were analyzed for major cations [calcium (Ca ²⁺), magnesium (Mg ²⁺), sodium (Na ⁺), potassium (K ⁺)], alkalinity, and trace metal concentrations.
Sampling strategy	The collected streamwater samples were either filtered onsite immediately or filtered in the lab within 24 h through 0.22 µm syringe filters, and divided into subsamples in acid-washed, filtered-streamwater-rinsed, high-density polyethylene (HDPE) bottles for major cations [calcium (Ca ²⁺), magnesium (Mg ²⁺), sodium (Na ⁺), potassium (K ⁺)], alkalinity, and trace metal analysis. The samples for trace metal analysis were acidified by adding 50% nitric acid. All the samples were stored at 4 °C until analysis. Sample sizes per 3-month lag ranged from 53–89 for pre-treatment groups (using WRTDS estimated data) and 9–25 for post-treatment groups (using measurement data). Despite smaller post-treatment counts, bootstrap resampling of the GAM-DiD model demonstrated stable treatment effect estimates with narrow 95% confidence intervals, confirming that replication was sufficient for robust statistical inference.
Data collection	All measurements were performed at the Yale Analytical and Stable Isotope Center (YASIC). The data was recorded by Fengchao Sun.
Timing and spatial scale	Streamwater samples were collected biweekly or monthly before basalt application from May 2022 to June 2023, because the streamwater chemistry was relatively stable. In order to monitor the streamwater chemistry changes after basalt application, streamwater samples were collected once to twice per week after basalt application from July 2023 to April 2025, with more intensive sampling during storms and snowmelt (hourly). There was a sampling gap between December 2022 to April 2023 because of the snowy and frozen conditions in the winter in VT. The stream weir location for discharge measurement, where water samples were also collected, is Lat: 44.45958 Lon: -72.0920.
Data exclusions	No data were excluded.
Reproducibility	Analytical measurement errors for duplicate water samples were generally within 3%. All data processing and statistical analyses are reproducible. Model scripts and data sources are fully described in the Methods section and/or provided in the Supplementary Materials.
Randomization	Randomization was not applied. Treatment was implemented at the watershed scale, and effects were evaluated using a quasi-experimental difference-in-differences framework based on predefined pre- and post-treatment periods.
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Field work, collection and transport

Field conditions	W-2 watershed is a low-elevation (285 m to 377 m) agricultural catchment, with 27% forest and 73% hayfield and pasture. The area experiences a cold, humid continental climate, with mean annual precipitation of 1,050 mm yr ⁻¹ and mean annual air temperature of 5.7 degree Celsius. The mean daily discharge from the W-2 stream averages 0.76 ± 0.19 mm d ⁻¹ .
Location	The study site, the W-2 watershed, is a 59-ha headwater catchment situated within the Sleepers River Research Watershed in Danville, Vermont, USA. The stream weir location for discharge measurement, where water samples were also collected, is Lat: 44.45958 Lon: -72.0920.
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☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☒	☐ Demonstrate how to render a vaccine ineffective
☒	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.
Files in database submission	Provide a list of all files available in the database submission.
Genome browser session (e.g. UCSC)	Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates	Describe the experimental replicates, specifying number, type and replicate agreement.
Sequencing depth	Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.
Antibodies	Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.
Peak calling parameters	Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.
Data quality	Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.
Software	Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

- Confirm that:
- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.
Instrument	Identify the instrument used for data collection, specifying make and model number.
Software	Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI

☐

Used

☐

Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See [Eklund et al. 2016](#))

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

- | | |
|--------------------------|---|
| n/a | Involvement in the study |
| ☐ | ☐ Functional and/or effective connectivity |
| ☐ | ☐ Graph analysis |
| ☐ | ☐ Multivariate modeling or predictive analysis |

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.